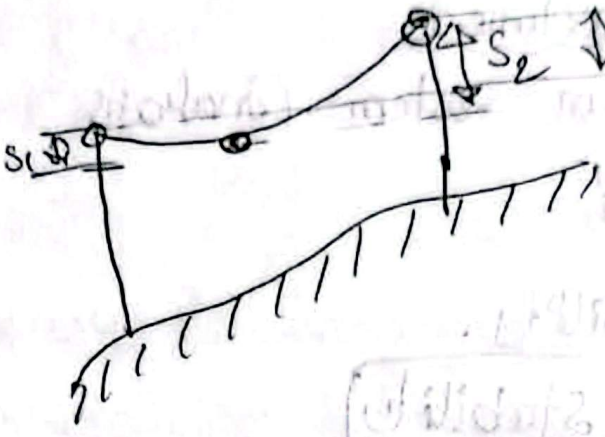
Subject: Lecture - 12 Date:



$$\therefore \omega t = \sqrt{(\omega_1 + \omega_2)^2 + (\omega_1 \omega)^2}$$

Seeg, $S = \frac{\omega t^2}{2T}$

CH - 08 → END here / presentation

Assignment + tut → End line

Article 8.1, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7, 8.8, 8.9, 8.10, 8.11, 8.12, 8.13, 8.14, 8.15

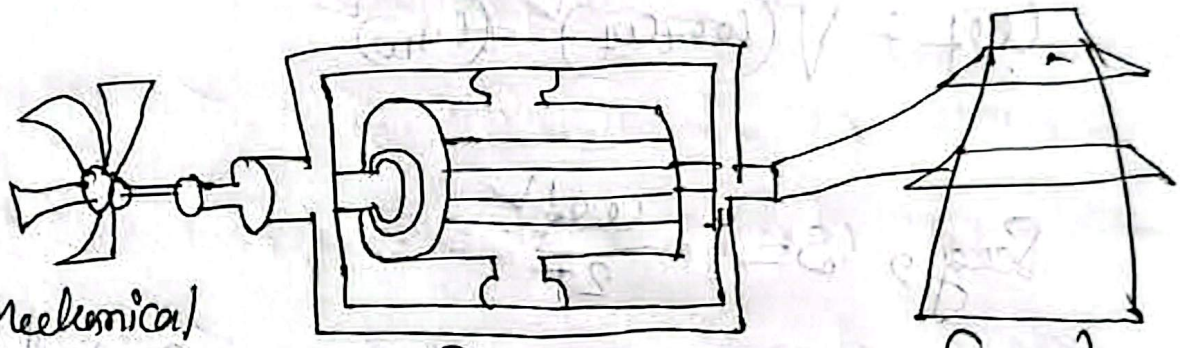
Subject: Date: 2nd

Lecture-18

Book: Modern Power System Analysis
D.P Kothari
I.J Nagrath

CH-12.
Power system Stability.

Stability



Kinetic Energy → Synchronous generator/ Alternator → Electrical Energy

$$KE = \frac{1}{2} J \omega_{sm}^2$$

ω_{sm} = Mechanical speed of Rotor
J = Moment of Inertia

Subject: Date :

If KE ↑ ω_{rm} ↑ Electrical speed ↑
KE ↓ ω_{rm} ↓ Electrical speed ↓

$N_s = \frac{120f}{P}$ ↓

Frequency is changing. For a synchronous machine frequency need to be constant.

If, ↑ side: Load ↑ Need more electrical energy.

KE ↓ ω_{rm} ↓ N_s ↓ f ↓

∴ Load ↑ frequency ↓ [Disturbance]

To solve this problem speed governor will give extra Kinetic energy to Rotor.

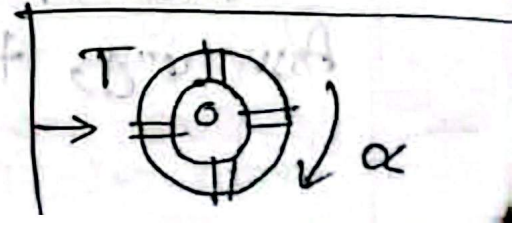
*** Rotor angle < power angle

Swing equation: It help us to model and predict how generates behave during unstable condition and faulty conditions.

Newtons law: $T = J \times \alpha$

T = Torque
J = Moment Inertia
α = Angular Acceleration

For generators rotor the change is speed depends on how much torque is acting and how much inertia the rotor has.



Subject: Date:

There are two types of torque acting the rotor.

① Mechanical Torque (T_m): that comes from the turbine trying to spin the rotor.

② Electrical Torque (T_e): that resists motor as the generator pushes power into grid.

Case 1: $T_m = T_e$; Rotor speed stays steady.

Case 2: $T_m > T_e$; Rotor speeds up.

Case 3: $T_m < T_e$; Rotor speeds down.

$\therefore T_{net} = T_m - T_e$. Here, δ = Rotor Angle

Applying Newton's law

$$\alpha \propto \frac{d^2\delta}{dt^2} = T_m - T_e$$

physical form of swing equation on Rotational motion.

* Power = P $P = T \times \omega$ = Torque × Angular speed

Assuming; the rotor speed doesn't vary too much,

Subject: Date :

$$\frac{d^2\delta}{dt^2} = \frac{P_m - P_e}{M}$$

Power

$$\Rightarrow \boxed{M \times \frac{d^2\delta}{dt^2} = P_m - P_e}$$

Basic swing equation in Power term.

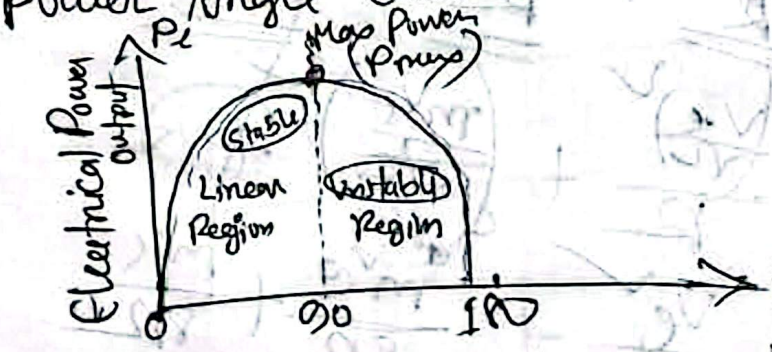
Different machines have different rating, sizes and inertia. To compare them easily we use inertia constant = H. Inertia Constant defines how much KE is stored per MVA of machine capacity.

using H

$$\boxed{\frac{2H}{\omega} \times \frac{d^2\delta}{dt^2} = P_m - P_e}$$

swing eqtn in per unit system.

Power Angle Curve



Pe depends on δ

$$P_e = \frac{EV}{X} \sin\delta$$

when δ = small, O/P increases almost linearly

(ii) Electrical torque (τ_e): That resists motor as the generator pushes power into the grid.

Case-1 : $\tau_m = \tau_e$; Rotor speeds stays steady.
Case-2 : $\tau_m > \tau_e$; Rotor speeds up
case-3 : $\tau_m < \tau_e$; Rotor slows down.
$\therefore \tau_{net} = \tau_m - \tau_e$; Here, δ = Rotor angle.

Applying Newton's law,

$\Rightarrow M \times \frac{d^2\delta}{dt^2} = \tau_m - \tau_e$

physical format solving equation on ab. Rotational motion.

Power = P $P = \tau \times \omega$ = torque × Angular speed

Assuming, Rotor speed doesn't vary too much,

$$\frac{d^2\delta}{dt^2} = \frac{P_m - P_e}{M}$$

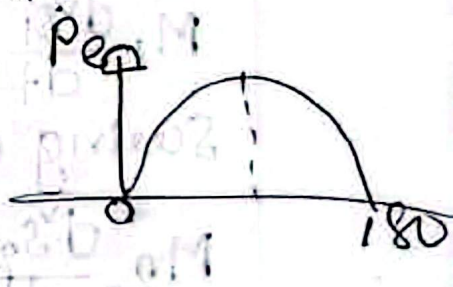
$$\Rightarrow M \times \frac{d^2\delta}{dt^2} = P_m - P_e$$

M = inertia

Basing swing eqtn in Power term.

Keeping δ

Keeping δ within safe limits is essential for grid stability.



Swing eqtn derivation in exam

Initial Condition must Consider

(ii) Electrical torque (τ_e): That resists motor as the generator pushes power into the grid.

Case-1 : $\tau_m = \tau_e$; Rotor speeds stays steady.
Case-2 : $\tau_m > \tau_e$; Rotor speeds up
case-3 : $\tau_m < \tau_e$; Rotor slows down.
$\therefore \tau_{net} = \tau_m - \tau_e$; Here, δ = Rotor angle.

Applying Newton's law,

$\Rightarrow M \times \frac{d^2\delta}{dt^2} = \tau_m - \tau_e$

physical format solving equation on ab. Rotational motion.

Power = P $P = \tau \times \omega$ = torque × Angular speed

Assuming, Rotor speed doesn't vary too much,

$$\frac{d^2\delta}{dt^2} = \frac{P_m - P_e}{M}$$

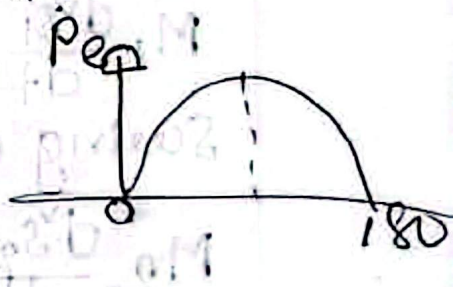
$$\Rightarrow M \times \frac{d^2\delta}{dt^2} = P_m - P_e$$

M = inertia

Basing swing eqtn in Power term.

Keeping δ

Keeping δ within safe limits is essential for grid stability.



Swing eqtn derivation in exam

Initial Condition must Consider

Leeture -14

Dynamics of a synchoronus machine

$$K.E = G_1 H = \frac{1}{2} M \omega_s \ (MJ)$$

Here, G_1 = Machine Rating in MVA/3-ϕ

M = Inertia Constant in MJ/MVA

$$M = \frac{2GH}{\omega_s} = \frac{GH}{\pi f} \quad MJ\text{-}s/\text{elect rad}$$

Swing equation of multiplicati machine system:

Multimachine system consists of a number of generator which are different ratings and connected with each other in parallal load angle is.

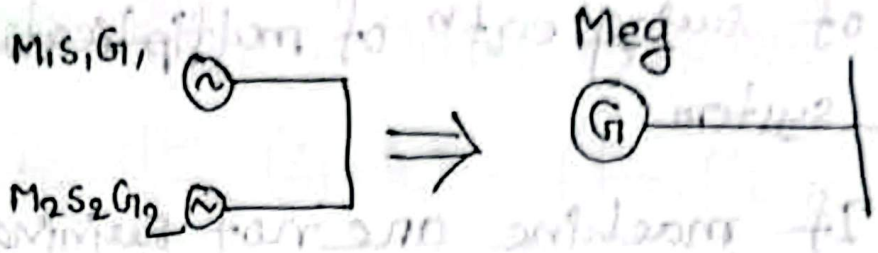
Angle of Momentum = M

Swing equation for machine 1(G_1)

$$M_1 \frac{d^2\delta_1}{dt^2} = P_{m1} - P_{e1} \quad \text{---(I)}$$

Swing equation for machine 2(G_2)

$$M_2 \frac{d^2\delta_2}{dt^2} = P_{m2} - P_{e2} \quad \text{---(II)}$$



If Machine are swinging co-herently ($S_1 = S_2$)

Adding equation ① and ②

$$\frac{d^2\delta}{dt^2}[M_1 + M_2] = P_{m1} - P_{e1} + P_{m2} - P_{e2}$$

$$\Rightarrow \frac{d^2\delta}{dt^2} M_{eq} = (P_{m1} + P_{m2}) - (P_{e1} + P_{e2})$$

$$= P_{m,eq} - P_{e,eq}$$

$$\frac{d^2\delta}{dt^2} M_{eq} = P_{m,eq} = P_{e,eq}$$

Here,

$$M_{eq} = M_1 + M_2$$

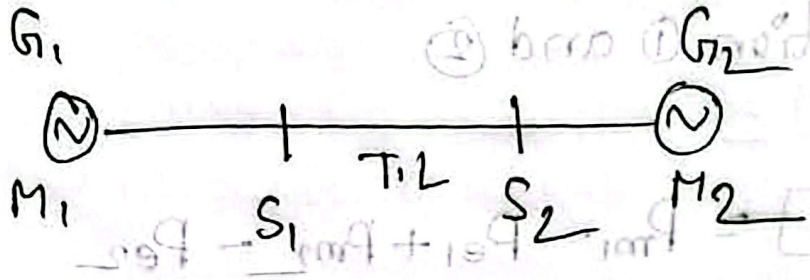
$$\Rightarrow \frac{G_{eq} H_{eq}}{f} = \frac{G_1 H_1}{f} + \frac{G_2 H_2}{f}$$

So,

$$\boxed{H_{eq} = \frac{G_1 H_1 + G_2 H_2}{G_{eq}}}$$

Formula of swing eqtn of multiplication machine system:

Case-2: If machine are not swining Co-herently. ($\delta_1 \neq \delta_2$)



Consider, $S_1 > S_2$

Real Power flow from higher rotor angle to lower rotor angle. As, $S_1 > S_2$ Real Power will flow from $G_1 \equiv G_2$

So, G_1 will act as generator (Providing the power G_2 will act as motor accepting the power)

And, $P_{e2} = -P_{e1}$ ——— ③

Multiplying eqtn ① by M_2 and eqtn ② by M_1.

$$M_1 M_2 \frac{d^2 \delta_1}{dt^2} = M_2 P_{m1} - M_2 P_{e1} \text{ ——— ④}$$

$$M_1 M_2 \frac{d^2 s_2}{dt^2} = M_1 P_{m2} - M_1 P_{e2} \quad \text{---} \textcircled{5}$$

$$M_1 M_2 \frac{d^2}{dt^2}(s_1 - s_2) = (P_{m1} M_2 - P_{m2} M_1) - (P_{e1} M_2 + P_{e2} M_1)$$

$$= (P_{m1} M_2 - P_{m2} M_1) - P_e [M_1 + M_2]$$

Divide both sides by $(M_1 + M_2)$

$$\frac{M_1 M_2}{M_1 + M_2} \frac{d^2 s}{dt^2} = \frac{P_{m1} M_2 - P_{m2} M_1}{M_1 + M_2} - P_{e1}$$

$$\Rightarrow M_{eq} = \frac{G H_{eq}}{\pi f} \Rightarrow \boxed{H_{eq} = \frac{M_{eq} \pi f}{G}}$$

derivation ☆☆☆

Equal area Criteria

Transient stability analysis of the simple system that can be ~~performed~~ performed without slowing any ~~eqt~~ is called as equal area criteria.

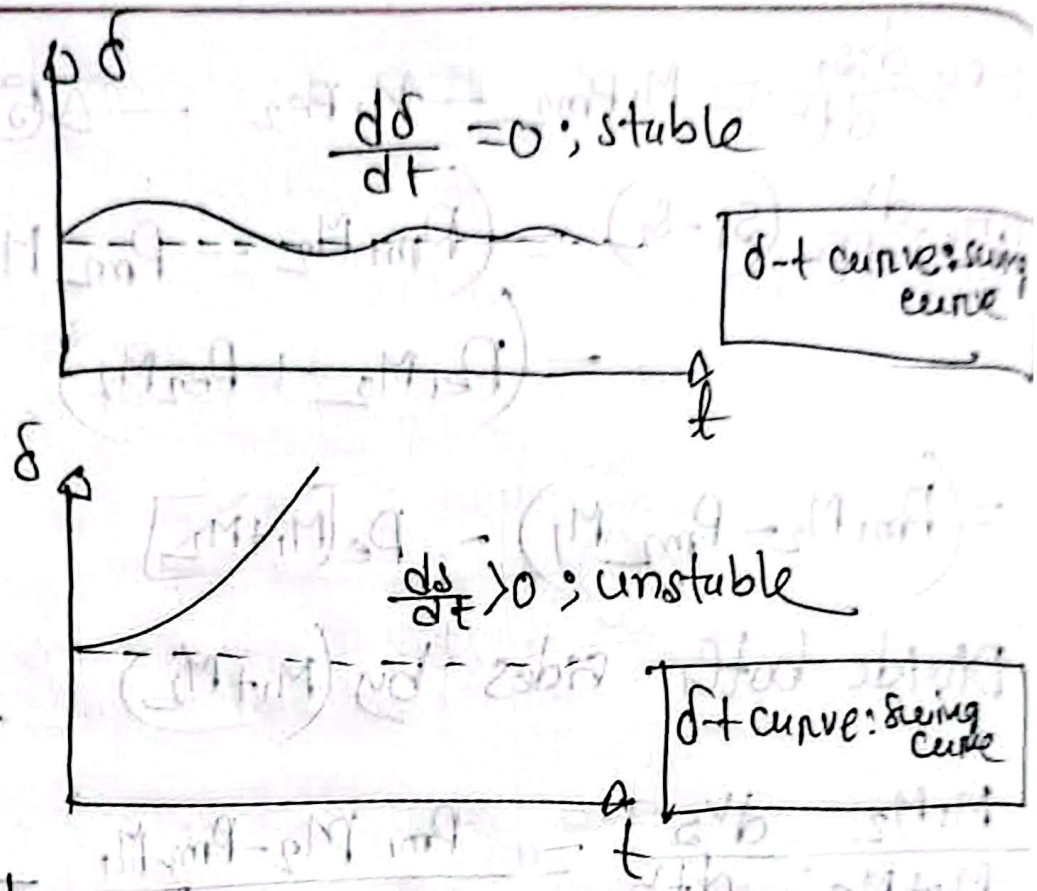


Fig:1

* If load angle (δ) between 2 machines is increased to a max value and then it is going to decrease or oscillation with a const. ampli. then it is considered as stable condition.

Fig:2

* If load angle δ is go on increasing continuously of a 2 machine system w/o any limit then the system is considered as unstable.

Advantage:

* This Method applied to any 2-machine system.

* For one machine system, swing with r. to on infinite bus can be analyzed by equal area criteria w/o using the numerical Method, which includes large calculation.

Disadvantage:

This method is not applicable for multimachine system.

Swing equation

$$M\frac{d^2\delta}{dt^2} = (P_m - P_e) \longrightarrow (1)$$

Consider, $P_a = P_m - P_e$

multiply both sides by $2\frac{d\delta}{dt}$

$$2M\frac{d\delta}{dt}\left(\frac{d^2\delta}{dt^2}\right) = 2\frac{d\delta}{dt}P_a \longrightarrow (2)$$

Integrating eqn 2 w.r.t t with the limits δ_0 to δ.

$$\Rightarrow \left(\frac{d\delta}{dt}\right)^2 = \int_{\delta_0}^{\delta} \frac{2}{M} P_a \, d\delta$$

As, $\int f(x) f'(x) dx = \frac{[f(x)]^2}{2}$

$$\Rightarrow \frac{d\delta}{dt} = \sqrt{\int_{\delta_0}^{\delta} \frac{2}{M} P_a \, d\delta}$$

Here, δ_0 = 1 load angle at which the machine seperating while turning at Synchronous speed under normal/stable condition.

For a stable system

$$\frac{d\delta}{dt} = 0$$

$$\therefore \int_{\delta_0}^{\delta} P_a \, d\delta = 0$$

$$\Rightarrow \int_{\delta_0}^{\delta} (P_m - P_e) \, d\delta = 0$$

$$\Rightarrow \int_{\delta_0}^{\delta} P_m \, d\delta = \int_{\delta_0}^{\delta} P_e \, d\delta$$

So, At δ_0 $\frac{d\delta}{dt} = 0$

Area under +ve Area

Area under -ve Area

Lecture-15

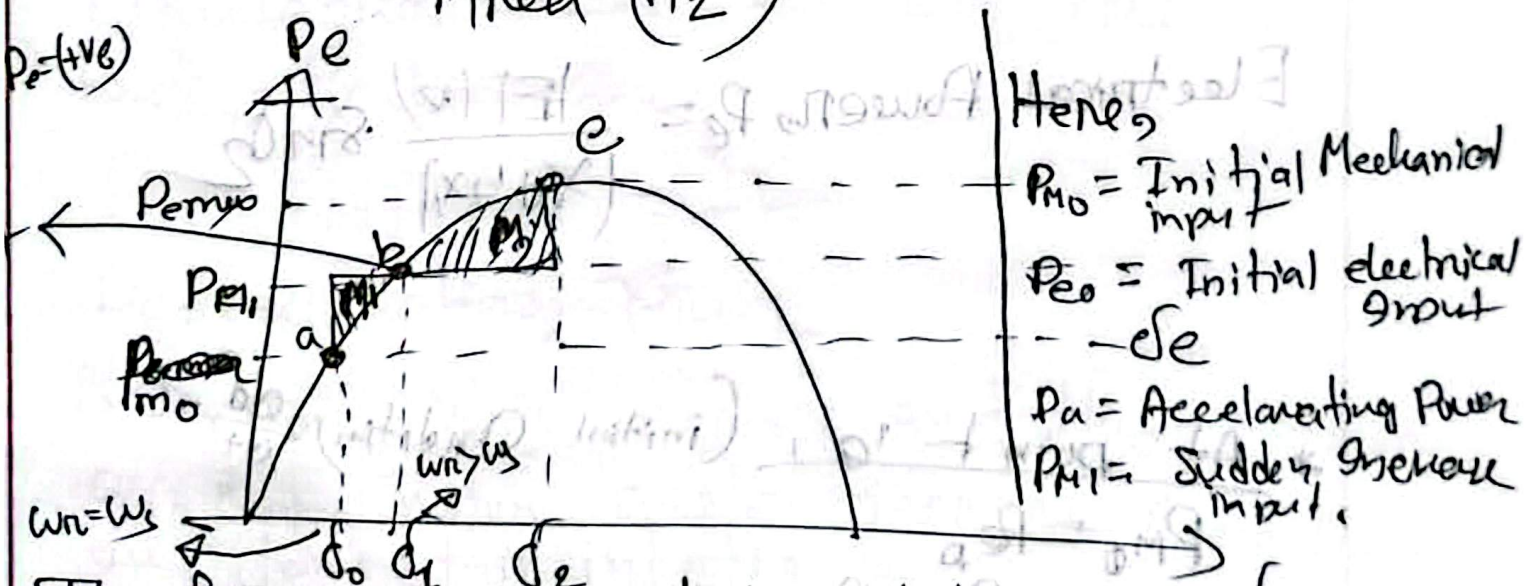
According to equal area criterion.

$$\int_{\delta_0}^{\delta} P_a \, d\delta = 0$$

$$\Rightarrow \int_{\delta_0}^{\delta} (P_m - P_e) \, d\delta = 0$$

If $P_m > P_e$. Acceleration occur Generator action positive area (A_1)

If $P_m < P_e$. Deacceleration, occurs Negative Area (A_2)



The following points are to be noted.

① There is no change in load angle (δ) when the rotor is exactly equal to syn. speed [$\delta = \delta_0$; $\omega = \omega_s$]

② The angle increases when $P_m > P_e$ [As $P_a \uparrow = P_m \uparrow - P_e$; $P_e = +ve$]

③ The angle decreases when $P_m > P_{en}$
As Φ $P_a = P_m - P_e \uparrow = -ve$

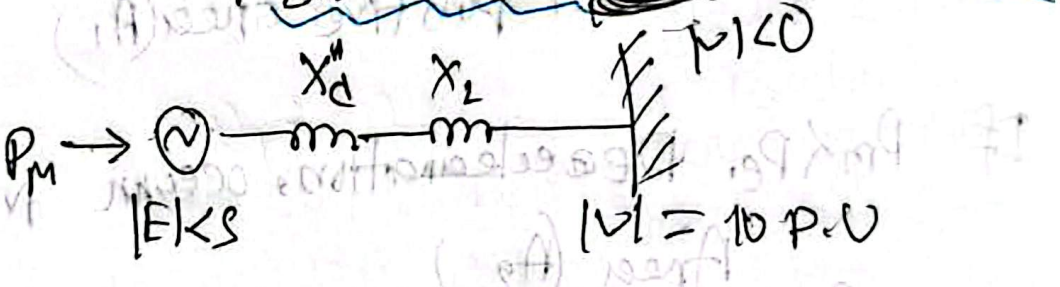
When,

① $M_1 = M_2$, Accelerating power $P_a = 0$
System stable.

$A_1 \rightarrow$ Kinetic energy stored by rotor swing oscillation.

$A_2 \rightarrow$ Kinetic energy given up by rotor to the system.

Sudden change in mechanical input



Electrical Power, $P_e = \frac{|E||V|}{|X_d' + X|} \sin\delta_2$

* At point '0' (initial condition) $\frac{d\delta}{dt} = 0$

$P_{M0} = P_{e0}$

$\therefore P_a = P_{M0} - P_{e0} = 0$

No accelerating power and $\omega_R = \omega_s$

No acceleration Power
and $\omega_R = \omega_s$

$\delta = \delta_0$

* From 'a' to 'b'

As mechanical input increases from P_{m0} to P_{m1}

$P_a = P_{m1} - P_e = +ve$

$P_m > P_e$

$P_a \uparrow$

At point 'b' to 'c'

Mechanical Input constant P_m, but P_e increases.

$P_a = P_{m1} - P_e \uparrow = -ve$

Decelerating Power

Now, ω_R starts reduce and this becomes ω_s. But rotor angle increase $\delta_1 \rightarrow \delta_2$. due to rotor inertia.

Accelerating power increases.

angle(δ) will increase from ($\delta_0 \rightarrow \delta_1$)

Rotor speed, will increase ($\omega_{rotor} > \omega_s$)

$\therefore P_m \uparrow P_a \uparrow \delta \uparrow \omega_r \uparrow$

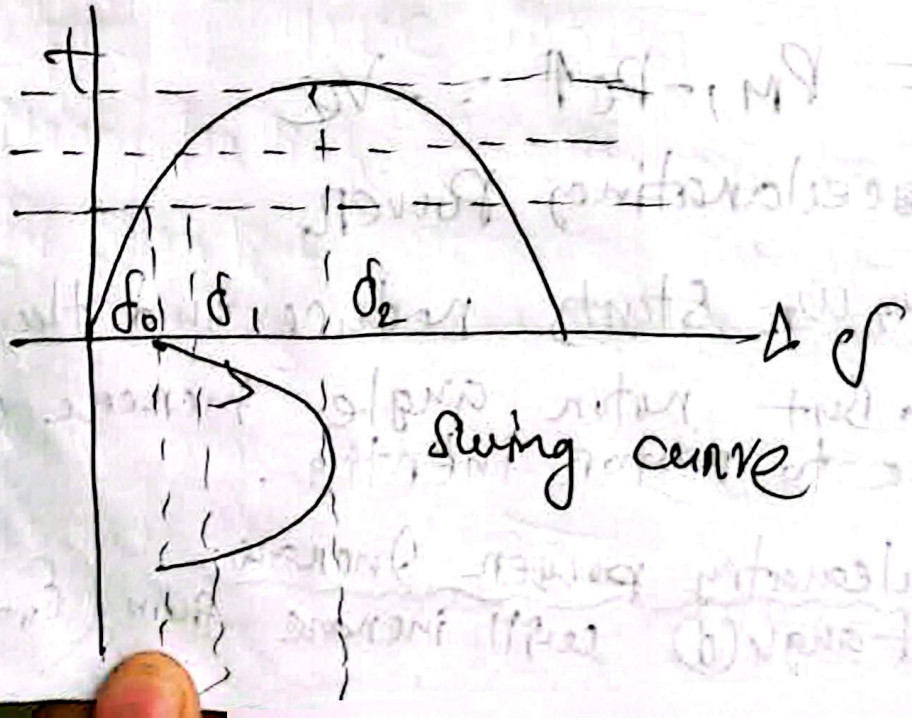
At point C

$\omega = \omega_s$

$P_{m1} < P_e$

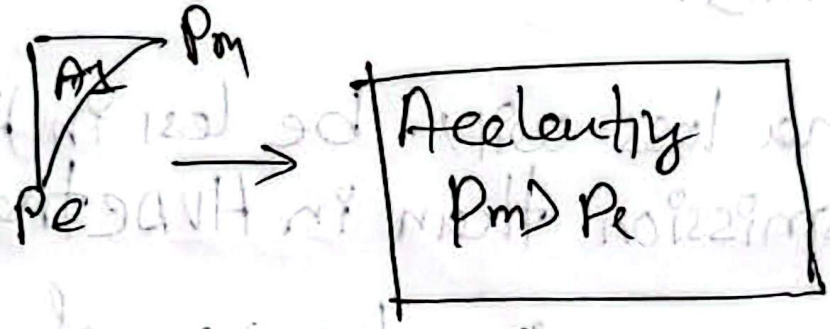
Here, Decelerating area and accelerating are both same

$A_1 = A_2$



$A_1 = \int_{\delta_0}^{\delta_1} (P_{M_1} - P_e)\, d\delta$; Accelarating

$A_2 = \int_{\delta_1}^{\delta_2} (P_e - P_{M_1})\, d\delta$, Deaccelating

* 

* 